

# Assignment: Multi-Agent Customer Support System using CrewAI

## 1. Objective

Design and implement a **hierarchical multi-agent customer support system** using CrewAI.

The system should:

- Automatically process customer queries
  - Classify and route them
  - Resolve using specialized agents
  - Use external search (Serper) when needed
  - Use a custom internal tool for knowledge retrieval
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## 2. System Architecture

### 2.1 Process Type

- `Process.hierarchical`

### 2.2 Hierarchy Structure

Support Manager (Top-Level Controller)

```
|  
|-- Ticket Classifier Agent  
|  
|-- Technical Support Agent  
|-- Billing Support Agent  
|-- General Support Agent  
|  
|-- Quality & Escalation Agent
```

The Manager is responsible for:

- Delegating tasks
  - Routing decisions
  - Final approval
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## 3. Agents Definition

### 3.1 Support Manager Agent

- Role: Supervisor and orchestrator
  - Responsibilities:
    - Receives user query
    - Assigns classification
    - Routes tasks
    - Handles escalation
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### 3.2 Ticket Classifier Agent

- Role: NLP classifier
  - Responsibilities:
    - Categorize query (technical, billing, general)
    - Detect priority and sentiment
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### 3.3 Technical Support Agent

- Role: Domain expert (technical issues)
- Responsibilities:
  - Debug errors
  - Provide solutions
  - Use Serper if needed

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### 3.4 Billing Support Agent

- Role: Payment and account specialist
  - Responsibilities:
    - Refunds
    - Payment failures
    - Subscription issues
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### 3.5 General Support Agent

- Role: FAQ handler
  - Responsibilities:
    - Answer general queries
    - Onboarding help
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### 3.6 Quality & Escalation Agent

- Role: Validator
  - Responsibilities:
    - Check correctness
    - Ensure tone and clarity
    - Trigger escalation
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## 4. Tools Integration

### 4.1 External Tool: Serper Search

Tool: SerperDevTool

Purpose:

- Fetch real-time information
- Help technical agent resolve unknown issues

Usage:

- Only invoked when internal knowledge is insufficient
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## 4.2 Custom Tool: Internal Knowledge Base

### Description

A custom tool that simulates a company knowledge base.

### Functionality

- Retrieve predefined solutions
- Map keywords to responses

### Example Design

Input: "refund policy"

Output: "Refunds are processed within 5-7 business days."

### Purpose

- Reduce dependency on external search
  - Provide fast, reliable answers
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## 5. Tasks Definition

### Task 1: Customer Query Intake

- Agent: Manager
- Input: Raw user query
- Output: Structured request

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## Task 2: Ticket Classification

- Agent: Classifier
- Output:

```
{  
  "category": "technical",  
  "priority": "high",  
  "sentiment": "negative"  
}
```

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## Task 3: Routing Decision

- Agent: Manager
  - Logic:
    - Technical → Technical Agent
    - Billing → Billing Agent
    - General → General Agent
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## Task 4: Query Resolution

- Agent: Domain-specific agent
  - Tools Used:
    - Internal Knowledge Tool
    - Serper (fallback)
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## Task 5: Response Generation

- Agent: Domain agent
- Output:
  - Clear, structured answer

- Human-like tone
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### **Task 6: Quality Check**

- Agent: QA Agent
  - Checks:
    - Accuracy
    - Completeness
    - Tone
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### **Task 7: Escalation Handling**

- Agent: Manager
  - Trigger Conditions:
    - Low confidence
    - Negative sentiment
    - Complex issue
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### **Task 8: Logging and Feedback**

- Store:
    - Query
    - Response
    - Outcome
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## **6. Execution Flow**

Step 1: User submits query

Step 2: Manager assigns classification task

Step 3: Classifier returns category

Step 4: Manager routes to correct agent

Step 5: Agent resolves using:

- Knowledge Base
- Serper (if needed)

Step 6: QA Agent validates response

Step 7: Manager approves or escalates

Step 8: Final response returned

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## 7. Sample Scenario

Input:

"My payment failed but money was deducted"

Flow:

- Classified as: Billing
- Routed to Billing Agent
- Knowledge tool used → Payment failure template
- QA checks tone
- Manager approves

Output:

We understand your concern. If the amount was deducted, it will be refunded within 5-7 business days. Please check your bank statement and contact support if the issue persists.

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## 8. Key Design Decisions

- Hierarchical control ensures reliability
- Tool separation avoids overuse of external APIs
- QA layer improves response quality
- Manager-driven routing mimics real-world support teams